Variance-Reduction Techniques

Dr. Md Nasim Adnan

- One of the points we have tried to emphasize throughout this course is that simulations driven by random inputs will produce random output. Thus, appropriate statistical techniques applied to simulation output data are imperative if the results are to be properly analyzed, interpreted, and used.
- Since large-scale simulations may require great amounts of computer time and storage, appropriate statistical analyses (possibly requiring multiple replications of the model, for example) can become quite costly.

- Sometimes the cost of even a modest statistical analysis of the output can be so high that the precision of the results will be unacceptably poor. The analyst should therefore try to use any means possible to increase the simulation’s efficiency.
- In this chapter, we focus on statistical efficiency, as measured by the variances of the output random variables from a simulation.
- If we can somehow reduce the variance of an output random variable of interest (such as average delay in queue or average cost per month in an inventory system) without disturbing its expectation, we can obtain greater precision.

- Sometimes if a Variance Reduction Technique (VRT), properly applied, can make the difference between an impossibly expensive simulation project and a frugal, useful one.
- Almost all VRTs require some extra effort on the part of the analyst (if only to understand the technique) and this, as always, must be considered.
- Almost all VRTs themselves will increase computing cost, and this decrease in computational efficiency must be traded off against the potential gain in statistical efficiency.

COMMON RANDOM NUMBERS

- The first VRT we consider, common random numbers (CRN), is actually different from the others in that it applies when we are comparing two or more alternative system configurations instead of investigating a single configuration.
- Despite its simplicity, CRN is the most useful and popular VRT of all the default setup in most simulation packages is that the same random-number streams and seeds are used unless specified otherwise. Thus, by default, the two configurations would actually use the same random numbers, though not necessarily properly synchronized.

ANTITHETIC VARIATES

- Antithetic variates (AV) is a VRT that is applicable to simulating a single system. As in CRN, we try to induce correlation between separate runs, but now we seek negative correlation.
- In its simplest form, AV tries to induce this negative correlation by using complementary random numbers to drive the two runs in a pair. That is, if U_k is a particular random number used for a particular purpose in the first run, we use $1 - U_k$ for this same purpose in the second run. It is valid to use $1 - U_k$ instead of simply a direct draw from the random-number generator, since $U \sim U(0, 1)$ implies that $1 - U \sim U(0, 1)$ as well.

CONTROL VARIATES

- Like CRN and AV, the method of control variates (CV) attempts to take advantage of correlation between certain random variables to obtain a variance reduction.
- In principle, at least, there is an appealing intuition to CV. Let X be an output random variable, such as the average of the first 100 customer delays in queue, and assume we want to estimate m = E(X). Suppose that Y is another random variable involved in the simulation that is thought to be correlated with X (either positively or negatively), and that we know the value of v = E(Y).
- For instance, Y could be the average of the service times of the first 99 customers who complete their service in the queueing model mentioned above, so we would know its expectation since we generated the service-time variates from some known input distribution.

- It is reasonable to suspect that larger-than-average service times (i.e., Y > v) tend to lead to longer-than-average delays (X > m) and vice versa; i.e., Y is correlated with X, in this case positively.
- Thus, if we run a simulation and notice that Y > v (which we can tell for sure since we know v), we might suspect that X is above its expectation m as well (although we would not know this for sure unless the correlation between Y and X were perfect), and accordingly adjust X downward by some amount.
- If it turned out, on the other hand, that Y < n, we would suspect that X < m as well and so adjust it upward instead. In this way, we use our knowledge of Y's expectation to pull X (down or up) toward its expectation m, thus reducing its variability about m from one run to the next.